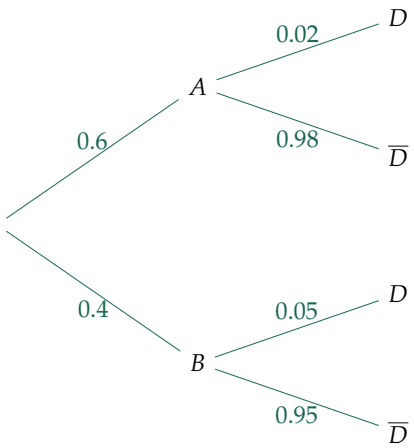
Teaching notes and solutions

> **Teacher note · Teacher notes.** Reactivate conditional probabilities in S36 before any Bernoulli scheme. Independence is an assumption of the model; it does not follow from repetition alone. For a draw without replacement, do not automatically invoke a binomial distribution. The formulas for expected value and variance are used in S39; their proof by a sum of indicators is revisited in Chapter 14. A threshold requested in December must be obtainable by a loop, as the logarithm has not yet been taught.

Solutions to the activity

1. The tree has branches A and B, with probabilities 0.6 and 0.4. From A, the branches $D, \overline{D}$ carry $0.02, 0.98$; from B, they carry $0.05, 0.95$.



   By total probability, $P(D) = P(A)P_A(D) + P(B)P_B(D) = 0.012 + 0.020 = 0.032$.

2. $P(D) > 0$ and $P(A \cap D) = 0.6 \times 0.02 = 0.012$. So $P_D(A) = \frac{P(A \cap D)}{P(D)} = \frac{0.012}{0.032} = \frac{3}{8} = 0.375$.

3. Each selection is a trial with two outcomes, "defective" or "not defective", with the same probability 0.032 of a defect. Since the twenty selections are independent, $X \sim \mathcal{B}(20, 0.032)$.

4. $P(X = 0) = 0.968^{20} \simeq 0.5218$. By complement,

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.968^{20} - 20(0.032)0.968^{19} \simeq 0.1332.$$

   Finally $E(X) = 20 \times 0.032 = 0.64$. This expected value does not imply that a selection has a fractional number of defects.

5. A common cause linked to the batch could make the defects dependent. The binomial distribution and its calculations rely on the independence given in the model; this assumption would need to be re-examined for a different protocol.

E1. By total probability, $P(B) = 0.3 \times 0.8 + 0.7 \times 0.1 = 0.31$. Since $P(B) > 0$, $P_B(A) = \frac{P(A \cap B)}{P(B)} = \frac{0.24}{0.31} = \frac{24}{31}$.

E2. By the binomial formula, $P(X = 3) = \binom{8}{3}(\frac{1}{4})^3(\frac{3}{4})^5 = 56(\frac{1}{4})^3(\frac{3}{4})^5$. The parameters give $E(X) = \frac{8}{4} = 2$ and $V(X) = 8(\frac{1}{4})(\frac{3}{4}) = \frac{3}{2}$.

E3. The complement of $X \geq 4$ is $X \leq 3$, so $P(X \geq 4) = 1 - F(3)$. For the interval, we remove from $X \leq 5$ the values $X \leq 1$, so $P(2 \leq X \leq 5) = F(5) - F(1)$.

E4. No, in a standard pack containing four aces. Without replacement, obtaining an ace on the first draw changes the probability of obtaining one on the second: the trials are not independent. Moreover, the number of aces can never equal 5, whereas a binomial distribution with parameters 5 and $0 < p < 1$ gives this value a positive probability.

E5. The probability of at least one success is $1 - (\frac{1}{2})^n$. The condition is equivalent to $2^n \geq 100$. Since $2^6 = 64 < 100 \leq 128 = 2^7$, the first integer is 7. The probability is then $\frac{127}{128} \simeq 0.99219$.

Solutions to Homework 13

1. X counts the successes of twenty independent Bernoulli trials with parameter 0.8. So $X \sim \mathcal{B}(20, 0.8)$, $E(X) = 20 \times 0.8 = 16$ and $V(X) = 20 \times 0.8 \times 0.2 = 3.2$.

2. The hall runs short of seats if $X > 18$, that is $X = 19$ or $X = 20$. Since these events are mutually exclusive,

$$P(X > 18) = 20(0.8)^{19}(0.2) + (0.8)^{20} \simeq 0.069175.$$

   The risk is about 6.92 %.

3. The number of refusals R equals 0 for $X \leq 18$, 1 for $X = 19$ and 2 for $X = 20$. By the definition of expected value,

$$E(R) = P(X = 19) + 2P(X = 20) \simeq 0.0807045.$$

   The takings are $20 \times 10 = 200$ and the payments $30R$, so $G = 200 - 30R$. Thus $E(G) = 200 - 30E(R) \simeq 197.5789$ monetary units. The 30 already includes all payments linked to a refusal; one must not subtract a further 10.

4. With eighteen reservations, no refusal is possible and the takings are $18 \times 10 = 180$, whatever the number of people present. The expected profit with twenty reservations exceeds this amount by about 17.58 monetary units.

5. The arrivals of people from the same group may be dependent. Moreover, the satisfaction of the public and the consequences of a refusal are not summed up by the expected profit. The result quantifies a model; it is not enough on its own to dictate a decision.

Solutions to the textbook or past-paper work

The 100 choices are independent, with the same probability $p = 0.0714$. So $X \sim \mathcal{B}(100; 0.0714)$.

$$P(X \leq 7) = \sum_{k=0}^{7} \binom{100}{k} (0.0714)^k (0.9286)^{100-k} \simeq 0.577.$$

Finally, $E(X) = 7.14$ and $V(X) = 100 \times 0.0714 \times 0.9286 = 6.630204 \simeq 6.63$.